

Gaussian covariance of windowed bispectrum multipoles in the tripolar (Sugiyama) basis

The same formalism as for the power spectrum: local plane-parallel correlators, plane-wave and Legendre expansions, Wigner–Eckart, and a window function from pair counts

Derivation note, branch thecov2, October 2026. Conventions follow Sugiyama, Saito, Beutler & Seo (2019, 2020).

Summary. The Gaussian (*PPP*) covariance of the bispectrum multipoles $B_{\ell_1 \ell_2 L}(k_1, k_2)$ of a windowed survey can be derived exactly as the power-spectrum covariance was: each of the three two-point correlators that a Wick pairing produces is a windowed, locally anisotropic power spectrum; the shell integrals turn into Bessel averages; the Legendre weights of the estimators and of the anisotropic correlators turn into spherical harmonics; and the Wigner–Eckart theorem collapses the product of Gaunt coefficients onto rotational invariants. Two things are new. First, the three correlators sit at three points, so the window enters through its **three-point** function $\int d^3x W(x)W(x+r_1)W(x+r_2)$, weighted by an invariant combination of harmonics of the two separations $\hat{r}_1, \hat{r}_2$ and of the three lines of sight. Second, the third side of the triangle, $k_3 = |k_1 + k_2|$, is not binned, so its power spectrum depends on the opening angle of the two binned momenta; this couples the two separations through a bipolar expansion. The six pairings split into two classes: in the two pairings where the two line-of-sight galaxies pair with each other the window function is anchored at that galaxy and is a product of two pair sums per anchor — no triplet counting; in the other four pairings one shell momentum runs along the third side of the triangle. For the elements that matter most, the diagonal $(k_1, k_2) = (k'_1, k'_2)$, the first class is the leading term and the second is suppressed by $\sim \Delta k/k$. In a periodic box the result reduces to eq. (49) of Sugiyama et al. (2020). Nothing here is specific to B_{000} or B_{202} ; their index sets are listed at the end.

1 Conventions

We use the identities of the power-spectrum note: the addition theorem, the plane-wave expansion, the generalised Gaunt coefficient $G_{\ell_1 \dots \ell_n}^{m_1 \dots m_n} \equiv \int d\Omega \prod_a Y_{\ell_a m_a}$, and the product rule

$$\prod_{a=1}^n Y_{\ell_a m_a}(\hat{n}) = \sum_{\Lambda \mu} G_{\ell_1 \dots \ell_n \Lambda}^{m_1 \dots m_n \mu} \bar{Y}_{\Lambda \mu}(\hat{n}), \quad \prod_{a=1}^n \bar{Y}_{\ell_a m_a}(\hat{n}) = \sum_{\Lambda \mu} G_{\ell_1 \dots \ell_n \Lambda}^{m_1 \dots m_n \mu} Y_{\Lambda \mu}(\hat{n}). \quad (1)$$

Spherical harmonics are the standard $Y_{\ell m}$; Sugiyama et al. use $y_\ell^m = \sqrt{4\pi/(2\ell+1)} Y_{\ell m}$.

Fields and windows. One tracer, weighted field $F(x) = w(x)[n_g(x) - \alpha n_r(x)]$, Fourier transform $F(k) = \int d^3x e^{-ik \cdot x} F(x)$, so $F(-k) = \overline{F(k)}$. With $\omega(x) \equiv \bar{n}(x)w(x)$:

$$W(x) \equiv \omega^2(x) \quad (\text{pair window}), \quad S(x) \equiv (1 + \alpha) \bar{n}(x) \omega^2(x) \quad (\text{shot-noise window}), \quad (2)$$

$$I_3 \equiv \int d^3x \omega^3(x).$$

I_3 is the normalisation $I = \int d^3x \bar{n}^3 w^3$ of Sugiyama et al. (2019, eq. 37).

Shell averages. For a shell i of the wavenumber magnitude, $V_i = \int_i d^3q/(2\pi)^3$ and

$$\langle\langle f \rangle\rangle_i \equiv \frac{1}{V_i} \int_i \frac{d^3q}{(2\pi)^3} f(q) = \frac{\int_i q^2 dq \int \frac{d\Omega_q}{4\pi} f(q)}{\int_i q^2 dq}, \quad (3)$$

so that the angular part is a normalised average; nested averages are abbreviated, $\langle\langle f \rangle\rangle_{ij} \equiv \langle\langle\langle f \rangle\rangle_i\rangle_j$ and $\langle\langle f \rangle\rangle_{ijij'}$ for the four shells of a covariance element. As in the power-spectrum note, with P_L the multipoles of the model power spectrum,

$$\bar{j}_\lambda^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr)}{\int_i q^2 dq}, \quad P_{L\lambda}^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr) P_L(q)}{\int_i q^2 dq}. \quad (4)$$

Plane-wave shell averages. From the plane-wave expansion and the orthogonality of the harmonics,

$$\langle \langle e^{-iq \cdot r} \bar{Y}_{\ell m}(\hat{q}) \rangle \rangle_i = (-i)^\ell \bar{j}_\ell^{(i)}(r) \bar{Y}_{\ell m}(\hat{r}), \quad \langle \langle e^{+ip \cdot r} \bar{Y}_{\Lambda \mu}(\hat{p}) g(p) \rangle \rangle_i = i^\Lambda \bar{g}_\Lambda^{(i)}(r) \bar{Y}_{\Lambda \mu}(\hat{r}), \quad (5)$$

with $\bar{g}_\Lambda$ the Bessel-weighted radial average of g . The first is used for the momenta of the estimator whose covariance we take, the second for the primed one.

Tripolar basis and estimator (Sugiyama et al. 2019, eqs. 12–13 and 34–35). With $H_{\ell_1 \ell_2 L} = \begin{pmatrix} \ell_1 & \ell_2 & L \\ 0 & 0 & 0 \end{pmatrix}$ and $N_{\ell_1 \ell_2 L} = (2\ell_1 + 1)(2\ell_2 + 1)(2L + 1)$,

$$B(k_1, k_2, \hat{n}) = \sum_{\ell_1 \ell_2 L} B_{\ell_1 \ell_2 L}(k_1, k_2) \mathcal{S}_{\ell_1 \ell_2 L}(\hat{k}_1, \hat{k}_2, \hat{n}), \quad (6)$$

$$\mathcal{S}_{\ell_1 \ell_2 L}(\hat{a}, \hat{b}, \hat{n}) = \frac{1}{H_{\ell_1 \ell_2 L}} \sum_{m_1 m_2 M} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} y_{\ell_1}^{m_1}(\hat{a}) y_{\ell_2}^{m_2}(\hat{b}) y_L^M(\hat{n}),$$

$$B_{\ell_1 \ell_2 L}(k_1, k_2) = N_{\ell_1 \ell_2 L} H_{\ell_1 \ell_2 L}^2 \int \frac{d\Omega_1 d\Omega_2 d\Omega_n}{(4\pi)^3} \bar{\mathcal{S}}_{\ell_1 \ell_2 L} B. \quad (7)$$

It is convenient to write $\mathcal{S}$ through the unnormalised combination of ordinary harmonics,

$$\tilde{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{a}, \hat{b}, \hat{n}) \equiv \sum_{m_1 m_2 M} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} Y_{\ell_1 m_1}(\hat{a}) Y_{\ell_2 m_2}(\hat{b}) Y_{LM}(\hat{n}), \quad \mathcal{S} = \frac{(4\pi)^{3/2}}{H\sqrt{N}} \tilde{\mathcal{S}}. \quad (8)$$

The estimator attaches the line of sight to the third galaxy, which is the one not integrated over a shell:

$$\begin{aligned} \hat{B}_{\ell_1 \ell_2 L}(i, j) &= \frac{NH^2}{I_3} \left\langle \left\langle \int d^3x \bar{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{x}) e^{i(q_1+q_2) \cdot x} F(q_1) F(q_2) F(x) \right\rangle \right\rangle_{ij} \\ &= \frac{(4\pi)^{3/2} H \sqrt{N}}{I_3} \left\langle \left\langle \int d^3x \tilde{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{x}) e^{i(q_1+q_2) \cdot x} F(q_1) F(q_2) F(x) \right\rangle \right\rangle_{ij}. \end{aligned} \quad (9)$$

Written out, $e^{i(q_1+q_2) \cdot x} F(q_1) F(q_2) F(x) = \int d^3y_1 d^3y_2 F(y_1) F(y_2) F(x) e^{-iq_1 \cdot (y_1-x)} e^{-iq_2 \cdot (y_2-x)}$: the two shell momenta are the sides of the triangle from the line-of-sight galaxy at x to the galaxies at y_1, y_2 .

Unbiasedness. In the local approximation $\langle F(y_1) F(y_2) F(x) \rangle \approx \omega^3(x) \zeta(y_1 - x, y_2 - x; \hat{x})$, so $\langle \hat{B} \rangle = (NH^2/I_3) \int d^3x \omega^3 \langle \langle \int d\Omega_1 d\Omega_2 / (4\pi)^2 \bar{\mathcal{S}}(\hat{q}_1, \hat{q}_2, \hat{x}) B(q_1, q_2, \hat{x}) \rangle \rangle$, which is $B_{\ell_1 \ell_2 L}$ averaged over the shells and over the survey, because for fixed $\hat{n}$ the $(\hat{q}_1, \hat{q}_2)$ average of $\bar{\mathcal{S}}_{\ell_1 \ell_2 L} \mathcal{S}_{\ell'_1 \ell'_2 L'}$ is $\delta_{\ell_1 \ell'_1} \delta_{\ell_2 \ell'_2} \delta_{LL'} / (NH^2)$ (orthogonality of the 3- j symbols at fixed M). The window dilutes it by $\int \omega^3 / I_3 = 1$; with a pair-averaged window this ratio is the bispectrum analogue of I_k / norm .

2 Wick expansion

Write the covariance as $\langle \hat{B} \bar{\hat{B}}' \rangle_c$ with the primed estimator at shells (i', j') and multipoles $(\ell'_1 \ell'_2 L')$. Since $\overline{F(p)} = F(-p)$,

$$\overline{\hat{B}}'_{\ell'_1 \ell'_2 L'}(i', j') = \frac{(4\pi)^{3/2} H' \sqrt{N'}}{I_3} \left\langle \left\langle \int d^3x' \tilde{\mathcal{S}}_{\ell'_1 \ell'_2 L'}(\hat{p}_1, \hat{p}_2, \hat{x}') e^{-i(p_1+p_2) \cdot x'} F(-p_1) F(-p_2) F(x') \right\rangle \right\rangle_{i' j'} \quad (10)$$

Label the six fields $1 \equiv F(q_1)$, $2 \equiv F(q_2)$, $3 \equiv F(x)$, $1' \equiv F(-p_1)$, $2' \equiv F(-p_2)$, $3' \equiv F(x')$. The Gaussian part of the six-point function is a sum over pairings; those with a pair inside one estimator, e.g. $\langle F(q_1) F(q_2) \rangle$, are proportional to the window at $q_1 + q_2$ and only survive when $|q_1 + q_2|$ is below the window scale, a fraction $\sim (\Delta k/k)^2$ of the shells: we drop them, as the power-spectrum note drops the $k \approx 0$ terms. What remains are the $3! = 6$ pairings between $\{1, 2, 3\}$ and $\{1', 2', 3'\}$:

$$\begin{aligned}
&\text{class 1: } (11')(22')(33'), (12')(21')(33'); \\
&\text{class 2: } (11')(23')(32'), (12')(23')(31'), (13')(21')(32'), (13')(22')(31').
\end{aligned} \tag{11}$$

In class 1 the two line-of-sight galaxies pair with each other; in class 2 each pairs with a shell galaxy of the other estimator.

3 The correlators in the local approximation

Three kinds of two-point correlator occur. With $P(p; \hat{n}) = \sum_L \mathcal{L}_L(\hat{p} \cdot \hat{n}) P_L(p)$, a pair of Fourier fields,

$$\langle F(q) F(-p) \rangle = \int d^3 y d^3 y' e^{-iq \cdot y + ip \cdot y'} \langle F(y) F(y') \rangle \approx \sum_L \int d^3 y e^{-i(q-p) \cdot y} W(y) \mathcal{L}_L(\hat{p} \cdot \hat{y}) P_L(p), \tag{12}$$

exactly as in the power-spectrum note (window at y , anisotropy about $\hat{y}$, pair separation only in the Fourier kernel; the power spectrum is evaluated at the primed momentum). A Fourier field and a configuration-space field, whose position carries the phase $e^{-ip_3 \cdot x'}$ of the primed estimator and is integrated,

$$\int d^3 x' e^{-ip_3 \cdot x'} \langle F(q) F(x') \rangle \approx \int d^3 y e^{-i(q+p_3) \cdot y} W(y) P(p_3; \hat{y}), \tag{13}$$

obtained by writing $x' = y + u$ and $\int d^3 u e^{-ip_3 \cdot u} \xi(u; \hat{y}) = P(p_3; \hat{y})$; and the two configuration-space fields, with both estimator phases,

$$\int d^3 x' e^{i(q_1+q_2) \cdot x - i(p_1+p_2) \cdot x'} \langle F(x) F(x') \rangle \approx W(x) e^{i(q_1+q_2-p_1-p_2) \cdot x} P(p_1 + p_2; \hat{x}). \tag{14}$$

In every case the line-of-sight harmonic of the primed estimator, $Y_{L'M'}(\hat{x}')$, is moved to the position of the galaxy its partner sits at ($\hat{x}$ in Equation 14, $\hat{y}$ in Equation 13), as the Legendre weight of the estimator was moved in the power-spectrum note. Shot noise adds to each correlator the same expression with $W \rightarrow S$ and $P_L \rightarrow \delta_{L0}^K$; we restore it at the end.

4 Class 1: the two lines of sight pair together

4.1 Phases and the three-point window function

For the pairing $(11')(22')(33')$ the three correlators are Equation 12 for (q_1, p_1) at y_1 , Equation 12 for (q_2, p_2) at y_2 , and Equation 14 at x . The phases combine into

$$e^{-i(q_1-p_1) \cdot y_1} e^{-i(q_2-p_2) \cdot y_2} e^{i(q_1+q_2-p_1-p_2) \cdot x} = e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2-p_2) \cdot r_2}, \quad r_1 \equiv y_1 - x, \quad r_2 \equiv y_2 - x \tag{15}$$

each momentum pair rides a plane wave on one side of the triangle from the line-of-sight galaxy. Collecting everything,

$$\begin{aligned}
C_{\ell\ell'}^{(1)}(ij; i'j') &= \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \langle \langle \mathcal{I}^{(1)} \rangle \rangle_{ij i' j'}, \quad \mathcal{I}^{(1)} \equiv \int d^3 x d^3 r_1 d^3 r_2 W(x) W(x+r_1) W(x+r_2) \\
&\times e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2-p_2) \cdot r_2} \overline{\tilde{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{x})} \tilde{\mathcal{S}}_{\ell'_1 \ell'_2 L'}(\hat{p}_1, \hat{p}_2, \hat{x}) P(p_1; \hat{y}_1) P(p_2; \hat{y}_2) P(p_1 + p_2; \hat{x}),
\end{aligned} \tag{16}$$

with $\hat{y}_a = (x + r_a) / |x + r_a|$. This is the master formula of class 1. The second class-1 pairing, $(12')(21')(33')$, is the same expression with the primed labels exchanged, $(p_1, \ell'_1, i') \leftrightarrow (p_2, \ell'_2, j')$; for the diagonal elements with $i = j$ it adds to the first with equal weight.

Box limit. For a periodic box of volume V with a global line of sight, $W = 1$, $\hat{x} = \hat{n}$, $I_3 = V$, and $\int d^3 r_1 e^{-i(q_1-p_1) \cdot r_1} = (2\pi)^3 \delta_D(q_1 - p_1)$, whose shell average over p_1 is $\delta_{ii'}/V_i$. Hence

$$\begin{aligned}
C_{\text{box}}^{(1)} &= \frac{(4\pi)^3 HH' \sqrt{NN'}}{VV_i V_j} \delta_{ii'} \delta_{jj'} \left\langle \langle \bar{\mathcal{S}}_\ell \tilde{\mathcal{S}}_{\ell'} P(q_1) P(q_2) P(|q_1 + q_2|) \rangle \right\rangle_{ij} \\
&= \frac{NN' H^2 H'^2 V}{\tilde{N}_{\text{mode}(k_i)} \tilde{N}_{\text{mode}(k_j)}} \delta_{ii'} \delta_{jj'} \langle \langle \mathcal{S}_\ell \mathcal{S}_{\ell'} PPP \rangle \rangle_{ij},
\end{aligned} \tag{17}$$

using $(4\pi)^3 \bar{\mathcal{S}} \tilde{\mathcal{S}}' = HH' \sqrt{NN'} \mathcal{S} \mathcal{S}'$ and $V_i = \tilde{N}_{\text{mode}(k_i)}/V$ with $\tilde{N}_{\text{mode}} = 4\pi k^2 \Delta k V / (2\pi)^3$. This is the first term of eq. (49) of Sugiyama et al. (2020), with their top-hat $W(k, k')$ as $\delta_{ii'}$; the “5 perms.” there are the other five pairings. With shot noise restored, $P \rightarrow P + 1/\bar{n}$ in each factor, as in their $P^{(N)}$.

4.2 Harmonic expansion

Momenta of the unprimed estimator. $\bar{\mathcal{S}}_\ell(\hat{q}_1, \hat{q}_2, \hat{x})$ carries $\bar{Y}_{\ell_1 m_1}(\hat{q}_1) \bar{Y}_{\ell_2 m_2}(\hat{q}_2) \bar{Y}_{LM}(\hat{x})$ and nothing else depends on $\hat{q}_1, \hat{q}_2$, so by Equation 5

$$\left\langle \langle e^{-iq_1 \cdot r_1} \bar{Y}_{\ell_1 m_1}(\hat{q}_1) \rangle \right\rangle_i = (-i)^{\ell_1} \bar{j}_{\ell_1}^{(i)}(r_1) \bar{Y}_{\ell_1 m_1}(\hat{r}_1), \tag{18}$$

and likewise for q_2 . The unprimed estimator contributes only bin-averaged Bessel functions, with no power spectrum, as in the power-spectrum note.

The vertex term. The third correlator depends on $\hat{p}_1$ and $\hat{p}_2$ jointly, through $p_3 = p_1 + p_2$:

$$P(p_1 + p_2; \hat{x}) = \sum_{L_3} \frac{4\pi}{2L_3 + 1} \sum_{\gamma} \bar{Y}_{L_3 \gamma}(\hat{x}) [Y_{L_3 \gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|)]. \tag{19}$$

The bracket is a function of $(\hat{p}_1, \hat{p}_2)$ at fixed magnitudes that transforms as a rank- L_3 tensor; its bipolar expansion

$$\begin{aligned}
Y_{L_3 \gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|) &= \sum_{ab} c_{ab}^{L_3}(p_1, p_2) [Y_a(\hat{p}_1) \otimes Y_b(\hat{p}_2)]_{L_3 \gamma}, \\
[Y_a \otimes Y_b]_{L \gamma} &\equiv \sum_{\alpha \beta} \langle a \alpha b \beta | L \gamma \rangle Y_{a \alpha}(\hat{p}_1) Y_{b \beta}(\hat{p}_2),
\end{aligned} \tag{20}$$

has coefficients $c_{ab}^{L_3}(p_1, p_2) = \int d\Omega_1 d\Omega_2 \overline{[Y_a \otimes Y_b]_{L_3 \gamma}} Y_{L_3 \gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|)$ (independent of γ), a one-dimensional integral over the opening angle after the rotational symmetry is used. Parity gives $a + b$ even, and $|a - b| \leq L_3 \leq a + b$. For $L_3 = 0$ the coefficients are the Legendre coefficients of $P_0(|p_1 + p_2|)$ in $\hat{p}_1 \cdot \hat{p}_2$: $a = b$ and $c_{aa}^0 \propto p_a(p_1, p_2)$ where $P_0(|p_1 + p_2|) = \sum_a p_a(p_1, p_2) \mathcal{L}_a(\hat{p}_1 \cdot \hat{p}_2)$. This expansion is the one new ingredient of the three-point case: it is finite in practice (the function is smooth in the opening angle; a few tens of terms resolve the BAO), and it is the only place where the two legs are coupled before the window is involved.

Momenta of the primed estimator. On leg 1 the factors of $\hat{p}_1$ are $Y_{\ell'_1 m'_1}(\hat{p}_1)$ from $\tilde{\mathcal{S}}_{\ell'}$, $Y_{L_1 M_1}(\hat{p}_1)$ from $\mathcal{L}_{L_1}(\hat{p}_1 \cdot \hat{y}_1) = (4\pi/(2L_1 + 1)) \sum_{M_1} Y_{L_1 M_1}(\hat{p}_1) \bar{Y}_{L_1 M_1}(\hat{y}_1)$, $Y_{a \alpha}(\hat{p}_1)$ from Equation 20, and the plane wave $e^{ip_1 \cdot r_1}$. Collapsing the three harmonics with Equation 1 and applying Equation 5,

$$\left\langle \langle e^{ip_1 \cdot r_1} Y_{\ell'_1 m'_1} Y_{L_1 M_1} Y_{a \alpha}(\hat{p}_1) g(p_1) \rangle \right\rangle_{i'} = \sum_{\Lambda \mu} G_{\ell'_1 L_1 a \Lambda}^{m'_1 M_1 \alpha \mu} i^\Lambda \bar{g}_\Lambda^{(i')}(r_1) \bar{Y}_{\Lambda \mu}(\hat{r}_1), \tag{21}$$

and the same on leg 2 with $(\ell'_2, m'_2, L_2, M_2, b, \beta, \Lambda', \mu', r_2)$. Because the vertex coefficient $c_{ab}^{L_3}(p_1, p_2)$ depends on both magnitudes, the two radial shell averages do not factorise; they form one two-dimensional kernel,

$$\mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')}(r_1, r_2) \equiv \frac{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2 j_\Lambda(p_1 r_1) j_{\Lambda'}(p_2 r_2) P_{L_1}(p_1) P_{L_2}(p_2) c_{ab}^{L_3}(p_1, p_2)}{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2}, \tag{22}$$

the analogue of $P_{L \Lambda}^{(i)}(s)$, now with the third power spectrum folded in through $c_{ab}^{L_3}$.

Harmonics at the five directions. After these steps the integrand of Equation 16 is a product of harmonics of $\hat{r}_1, \hat{r}_2, \hat{y}_1, \hat{y}_2, \hat{x}$:

$$\hat{r}_1 : \bar{Y}_{\ell_1 m_1} \bar{Y}_{\Lambda \mu} = \sum_{\lambda_1 \rho_1} G_{\ell_1 \Lambda \lambda_1}^{m_1 \mu \rho_1} Y_{\lambda_1 \rho_1}(\hat{r}_1), \quad \hat{r}_2 : \bar{Y}_{\ell_2 m_2} \bar{Y}_{\Lambda' \mu'} = \sum_{\lambda_2 \rho_2} G_{\ell_2 \Lambda' \lambda_2}^{m_2 \mu' \rho_2} Y_{\lambda_2 \rho_2}(\hat{r}_2), \quad (23)$$

$$\hat{y}_1 : \bar{Y}_{L_1 M_1}(\hat{y}_1), \quad \hat{y}_2 : \bar{Y}_{L_2 M_2}(\hat{y}_2), \quad \hat{x} : \bar{Y}_{LM} Y_{L' M'} \bar{Y}_{L_3 \gamma} = (-1)^{M'} \sum_{\Lambda_3 \mu_3} G_{LL' L_3 \Lambda_3}^{M, -M', \gamma, \mu_3} Y_{\Lambda_3 \mu_3}(\hat{x}). \quad (24)$$

The phase $(-i)^{\ell_1 + \ell_2} i^{\Lambda + \Lambda'}$ is real: $\Lambda + \Lambda' \equiv a + b \equiv 0 \pmod{2}$.

4.3 Wigner–Eckart and the invariant window function

The coefficient of $Y_{\lambda_1 \rho_1}(\hat{r}_1) Y_{\lambda_2 \rho_2}(\hat{r}_2) \bar{Y}_{L_1 M_1}(\hat{y}_1) \bar{Y}_{L_2 M_2}(\hat{y}_2) Y_{\Lambda_3 \mu_3}(\hat{x})$ is the tensor

$$\begin{aligned} T^{\rho_1 \rho_2 M_1 M_2 \mu_3} = & \sum_{m_1 m_2 M m'_1 m'_2 M' \alpha \beta \gamma \mu \mu'} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} \begin{pmatrix} \ell'_1 & \ell'_2 & L' \\ m'_1 & m'_2 & M' \end{pmatrix} \langle a \alpha b \beta \mid L_3 \gamma \rangle \\ & \times G_{\ell'_1 L_1 a \Lambda}^{m'_1 M_1 \alpha \mu} G_{\ell'_2 L_2 b \Lambda'}^{m'_2 M_2 \beta \mu'} G_{\ell_1 \Lambda \lambda_1}^{m_1 \mu \rho_1} G_{\ell_2 \Lambda' \lambda_2}^{m_2 \mu' \rho_2} (-1)^{M'} G_{LL' L_3 \Lambda_3}^{M, -M', \gamma, \mu_3}, \end{aligned} \quad (25)$$

built from Gaunt coefficients only, hence rotationally invariant. With five free magnetic indices the space of invariants is no longer one-dimensional: a coupling scheme is needed. The natural one follows the geometry — leg 1 ($\hat{r}_1, \hat{y}_1$), leg 2 ($\hat{r}_2, \hat{y}_2$), and the anchor $\hat{x}$:

$$E_{K_1 K_2}^{\rho_1 M_1 \rho_2 M_2 \mu_3} \equiv \sum_{\kappa_1 \kappa_2} \langle \lambda_1 \rho_1 L_1 M_1 \mid K_1 \kappa_1 \rangle \langle \lambda_2 \rho_2 L_2 M_2 \mid K_2 \kappa_2 \rangle \langle K_1 \kappa_1 K_2 \kappa_2 \mid \Lambda_3 \mu_3 \rangle, \quad (26)$$

in terms of which $T = \sum_{K_1 K_2} \mathcal{T}_{K_1 K_2} E_{K_1 K_2}$ with $\mathcal{T}_{K_1 K_2} = (\sum T E_{K_1 K_2}) / (\sum E_{K_1 K_2}^2)$ (sums over all five indices; the E for different (K_1, K_2) are orthogonal). The invariant window function is the three-point correlation of the window weighted by the corresponding scalar,

$$\begin{aligned} \mathcal{Q}^{\lambda_1 L_1 K_1; \lambda_2 L_2 K_2; \Lambda_3}(r_1, r_2) \equiv & \int d^3 x \int d^2 \hat{r}_1 d^2 \hat{r}_2 W(x) W(x + r_1) W(x + r_2) \\ & \times \sum_{\rho_1 M_1 \rho_2 M_2 \mu_3} E_{K_1 K_2}^{\rho_1 M_1 \rho_2 M_2 \mu_3} Y_{\lambda_1 \rho_1}(\hat{r}_1) \bar{Y}_{L_1 M_1}(\hat{y}_1) Y_{\lambda_2 \rho_2}(\hat{r}_2) \bar{Y}_{L_2 M_2}(\hat{y}_2) Y_{\Lambda_3 \mu_3}(\hat{x}), \end{aligned} \quad (27)$$

the analogue of $\mathcal{Q}_{\Lambda_1 \Lambda_2 \Lambda}(s)$ with one more separation and one more line of sight. It depends on the geometry only.

4.4 Result for class 1

$$\begin{aligned} C_{\ell \ell'}^{(1)}(ij; i' j') = & \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \sum_{L_1 L_2 L_3} \frac{(4\pi)^3}{(2L_1 + 1)(2L_2 + 1)(2L_3 + 1)} \sum_{ab} \sum_{\Lambda \Lambda'} \sum_{\lambda_1 \lambda_2 \Lambda_3} \sum_{K_1 K_2} (-1)^{(\Lambda + \Lambda' - \ell_1 - \ell_2)/2} \mathcal{T}_{K_1 K_2} \\ & \times \int r_1^2 dr_1 \int r_2^2 dr_2 \bar{j}_{\ell_1}^{(i)}(r_1) \bar{j}_{\ell_2}^{(j)}(r_2) \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')}(r_1, r_2) \mathcal{Q}^{\lambda_1 L_1 K_1; \lambda_2 L_2 K_2; \Lambda_3}(r_1, r_2), \end{aligned} \quad (28)$$

where $\mathcal{T}$ carries all the multipole labels $(\ell_1 \ell_2 L; \ell'_1 \ell'_2 L'; L_1 L_2 L_3; ab; \Lambda \Lambda'; \lambda_1 \lambda_2 \Lambda_3)$ and is a pure number computed once from 3- j symbols. The selection rules are those of the Gaunt coefficients: $\Lambda \in \text{tri}(\ell'_1, L_1, a)$, $\lambda_1 \in \text{tri}(\ell_1, \Lambda)$, $K_1 \in \text{tri}(\lambda_1, L_1)$, $\Lambda_3 \in \text{tri}(L, L', L_3) \cap \text{tri}(K_1, K_2)$, and the same for leg 2. Every sum is finite except those over (a, b) , which are the Legendre orders of the third power spectrum in the opening angle.

Shot noise. Each of the three correlators is linear in its window-times-spectrum, so restoring $W P_L \rightarrow W P_L + S \delta_{L_0}^K$ at each of the three points gives eight terms: at a leg, S at that point with $L_a = 0$ and $P_{L_a} \rightarrow 1$ inside Equation 22; at the vertex, $S(x)$ with $L_3 = 0$ and $c_{ab}^0 \rightarrow \delta_{a0} \delta_{b0} \sqrt{4\pi}$ (the constant $Y_{00} P_0 \rightarrow Y_{00}$). The window functions become $\mathcal{Q}^{XYZ}$ with $X, Y, Z \in \{W, S\}$ for the anchor and the two legs. The estimator's own shot-noise subtraction removes the self-pairs inside each estimator and leaves this structure intact, as it does for the power spectrum (Sugiyama et al. 2020, sec. 2.3).

5 Class 2: a line of sight pairs with a shell galaxy

For the pairing (11')(23')(32') the correlators are Equation 12 for (q_1, p_1) at y_1 , Equation 13 for (q_2, p_3) at y_2 with $p_3 = p_1 + p_2$, and Equation 12 read backwards for $(x, -p_2)$: $\int d^3 y'_2 e^{ip_2 \cdot y'_2} \langle F(x) F(y'_2) \rangle = e^{ip_2 \cdot x} W(x) P(p_2; \hat{x})$. The line-of-sight harmonic of the primed estimator now sits at $\hat{y}_2$. The phases combine into

$$e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2+p_1+p_2) \cdot r_2} = e^{-iq_1 \cdot r_1} e^{-iq_2 \cdot r_2} e^{-ip_2 \cdot r_2} e^{-ip_1 \cdot (r_2-r_1)}, \quad (29)$$

and the master formula is

$$C_{\ell\ell'}^{(2)} = \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \langle \langle \mathcal{I}^{(2)} \rangle \rangle_{ij i' j'}, \quad \mathcal{I}^{(2)} \equiv \int d^3 x d^3 r_1 d^3 r_2 W(x) W(x+r_1) W(x+r_2) \times e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2+p_1+p_2) \cdot r_2} \bar{\mathcal{S}}_\ell(\hat{q}_1, \hat{q}_2, \hat{x}) \tilde{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{y}_2) P(p_1; \hat{y}_1) P(p_1+p_2; \hat{y}_2) P(p_2; \hat{x}). \quad (30)$$

The other three class-2 pairings follow by the two relabellings $(p_1, \ell'_1, i') \leftrightarrow (p_2, \ell'_2, j')$ and $(q_1, \ell_1, i, r_1) \leftrightarrow (q_2, \ell_2, j, r_2)$.

Box limit. $\int d^3 r_1 d^3 r_2$ gives $(2\pi)^6 \delta_D(q_1 - p_1) \delta_D(q_2 + p_1 + p_2)$: the shell of q_2 must contain the third side of the primed triangle. With the shell averages,

$$C_{\text{box}}^{(2)} = \frac{(4\pi)^3 H H' \sqrt{N N'}}{V V_i V_j} \delta_{ii'} \times \langle \langle \mathbb{1}[|p_1 + p_2| \in j] \bar{\mathcal{S}}_\ell(\hat{p}_1, -\hat{p}_3, \hat{n}) \tilde{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{n}) \times P(p_1) P(p_2) P(|p_1 + p_2|) \rangle \rangle_{ij'}, \quad (31)$$

the terms of eq. (49) of Sugiyama et al. (2020) in which k_3 falls in a bin. On the diagonal $j' = j$ the indicator selects the configurations with $|p_1 + p_2| \in j$, a fraction $\sim \Delta k / (2k)$ of the shell of p_2 (the third side ranges over $|p_1 - p_2| \leq p_3 \leq p_1 + p_2$, a width $\sim 2k$): class 2 is suppressed by $\sim \Delta k / k$ on the diagonal, 5% at $k = 0.1$ for $\Delta k = 0.005$, and it is the only Gaussian contribution off the diagonal, where it is of the same relative size.

Structure. In the anchored basis the two plane waves of p_1 factorise, $e^{-ip_1 \cdot (r_2-r_1)} = e^{ip_1 \cdot r_1} e^{-ip_1 \cdot r_2}$, so the $\hat{p}_1$ average couples harmonics of $\hat{r}_1$ and $\hat{r}_2$:

$$\int \frac{d\Omega_{p_1}}{4\pi} e^{ip_1 \cdot r_1} e^{-ip_1 \cdot r_2} [Y \dots (\hat{p}_1)] = \sum_{\lambda \lambda'} i^{\lambda-\lambda'} j_\lambda(p_1 r_1) j_{\lambda'}(p_1 r_2) \times \int \frac{d\Omega_{p_1}}{4\pi} Y_\lambda(\hat{p}_1) \bar{Y}_{\lambda'}(\hat{p}_1) [Y \dots (\hat{p}_1)] \bar{Y}_\lambda(\hat{r}_1) Y_{\lambda'}(\hat{r}_2). \quad (32)$$

The Gaunt coefficient bounds $|\lambda - \lambda'|$ by the rank of the other harmonics of $\hat{p}_1$, but not λ itself: the expansion in harmonics of the two anchored separations does not terminate (it is the Bessel addition theorem for $j_\lambda(p_1 |r_2 - r_1|)$, which converges once $\lambda > p_1 \max(r_1, r_2)$). The rest of the derivation is as in class 1, with $\mathcal{Q}$ functions whose line-of-sight labels sit at $\hat{x}$ for L and at $\hat{y}_2$ for L' . Which orders are needed in practice is decided by the window, not by the kernel: see the next section.

6 Computing the window function from pair sums

The scalar in Equation 27 couples harmonics of $(\hat{r}_1, \hat{y}_1)$ into K_1 , of $(\hat{r}_2, \hat{y}_2)$ into K_2 , and the two into Λ_3 at the anchor. Therefore

$$\mathcal{Q}^{\dots}(r_1, r_2) = \sum_{\kappa_1 \kappa_2 \mu_3} \langle K_1 \kappa_1 K_2 \kappa_2 | \Lambda_3 \mu_3 \rangle \int d^3 x W(x) Y_{\Lambda_3 \mu_3}(\hat{x}) A_{K_1 \kappa_1}^{(1)}(x; r_1) A_{K_2 \kappa_2}^{(2)}(x; r_2), \quad (33)$$

$$A_{K\kappa}^{(a)}(x; r) \equiv \int d^2 \hat{r} W(x+r) \sum_{\rho M} \langle \lambda_a \rho L_a M | K \kappa \rangle Y_{\lambda_a \rho}(\hat{r}) \bar{Y}_{L_a M}(\hat{y}), \quad (34)$$

and $A_{K\kappa}^{(a)}(x; r)$ is a **pair** quantity: the harmonic moments of the window on the shell of radius r around the anchor x , accumulated from the pairs $(x, x + r)$ exactly as the pair counts of `WindowLibrary`, in bins of r . The three-point function is then the sum over anchors of the product of two such moments. This is the Slepian–Eisenstein construction of the three-point function, and it needs no triplets: the cost is (anchors) $\times$ (neighbours per anchor) for the moments, plus (anchors) $\times$ (bins of r_1) $\times$ (bins of r_2) $\times$ (label pairs) for the products. The same construction holds in class 2, where the two plane waves of p_1 are each attached to one anchored side.

What bounds the ranks. The exact expansions contain unbounded sums: over (a, b) in class 1 through $\lambda_{1,2} \in \text{tri}(\ell, \Lambda)$, and over (λ, λ') in class 2. The product of a kernel of rank λ in $\hat{r}_1$ with $\mathcal{Q}$ integrates to zero unless $\mathcal{Q}$ has that rank, and $\mathcal{Q}$ is smooth in the opening angle of the triangle: the window varies on hundreds of Mpc/h, so $\mathcal{Q}^\cdots(r_1, r_2)$ at ranks beyond $\lambda \sim 6\text{--}8$ is negligible for r_1, r_2 below the window scale, and exactly zero in a periodic box, where only the fully isotropic component $\lambda_1 = \lambda_2 = \Lambda_3 = L_a = 0$ survives and reproduces the box formulae above (the isotropic component of the kernel is the full angular average, which is what the delta functions select). Truncating the window’s ranks at λ_w therefore truncates every sum, with an error set by the window’s angular smoothness and not by the kernels. The (a, b) sums of class 1 are then bounded by $\lambda_w + \ell + L_a$, and the class-2 (λ, λ') sums by λ_w ; alternatively the kernel’s projection on the low-rank basis is computed by direct angular quadrature of Equation 16 and Equation 30 at each (r_1, r_2) , which avoids the Gaunt algebra for the primed momenta altogether. The convergence in λ_w is checked by computing $\mathcal{Q}$ to a higher rank, at pair-count cost.

Cost. With $n_{\text{sub}} = 5000$ far randoms as anchors and neighbours: 2.5×10^7 pair evaluations for the moments (as now), and for the products $5000 \times n_r^2 \times (\text{label pairs}) \times (\text{magnetic contractions})$. For $\lambda_w = 4$, $L_a \leq 2$ and the diagonal elements the label pairs are ~ 2500 and the contraction $\sim 10^2$, i.e. $\sim 10^{11}$ operations per 100×100 radial grid: minutes on a GPU, an hour on a node. The near-pair regime ($s < s_{\text{split}}$) uses the tree neighbours as now. Rotating each anchor’s frame to $\hat{x} = \hat{z}$ makes $\mu_3 = 0$ and removes one magnetic sum.

7 B_{000} and B_{202} on the diagonal

B_{000} . $\ell_1 = \ell_2 = L = 0$ on both sides, $H = 1$, $N = 1$. The six Y_{00} factors supply $(4\pi)^{-3}$ and cancel the $(4\pi)^3$ of the prefactor. The $\hat{r}_1$ collapse forces $\lambda_1 = \Lambda$, the $\hat{r}_2$ one $\lambda_2 = \Lambda'$, and the anchor $\Lambda_3 = L_3$, so Equation 28 becomes

$$C_{000}^{(1)}(ij; i' j') = \frac{1}{I_3^2} \sum_{L_1 L_2 L_3} \frac{(4\pi)^3}{(2L_1 + 1)(2L_2 + 1)(2L_3 + 1)} \sum_{ab\Lambda\Lambda' K_1 K_2} (-1)^{(\Lambda + \Lambda')/2} \mathcal{T}_{K_1 K_2} \quad (35)$$

$$\times \int \int r_1^2 r_2^2 dr_1 dr_2 \bar{j}_0^{(i)}(r_1) \bar{j}_0^{(j)}(r_2) \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')} \mathcal{Q}^{\Lambda L_1 K_1; \Lambda' L_2 K_2; L_3},$$

with $\Lambda \in \text{tri}(L_1, a)$, $\Lambda' \in \text{tri}(L_2, b)$. In real space ($L_1 = L_2 = L_3 = 0$, $a = b$, $\Lambda = \Lambda' = K_1 = K_2 = a$) the scalar of Equation 27 is proportional to $\mathcal{L}_a(\hat{r}_1 \cdot \hat{r}_2)$: the window enters through its three-point function projected on Legendre polynomials of the opening angle, and the sum over a is the Legendre expansion of $P(|p_1 + p_2|)$; in the box only $a = 0$ survives and the kernel reduces to the angle-averaged $P(|q_1 + q_2|)$. In redshift space the ranks are $L_{1,2,3} \in \{0, 2, 4\}$, with $\Lambda_3 = L_3$: the window’s line-of-sight dependence at the anchor is the quadrupole and hexadecapole of the vertex correlator, and at the legs those of the two leg correlators.

B_{202} . $\ell_1 = 2$, $\ell_2 = 0$, $L = 2$ (and the same primed), $\mathcal{S}_{202}(\hat{k}_1, \hat{k}_2, \hat{n}) = \mathcal{L}_2(\hat{k}_1 \cdot \hat{n})$, $H_{202} = -\sqrt{1/5}$, $N = 25$. The 3- j symbols are $\begin{pmatrix} 2 & 0 & 2 \\ m_1 & 0 & -m_1 \end{pmatrix} = (-1)^{m_1}/\sqrt{5}$ and reduce the double sums to one magnetic index per estimator. The index sets: at $\hat{r}_2$ still $\lambda_2 = \Lambda'$ (because $\ell_2 = 0$); at $\hat{r}_1$, $\lambda_1 \in \text{tri}(2, \Lambda)$, with $\Lambda \in \text{tri}(2, L_1, a)$; at the anchor $\Lambda_3 \in \text{tri}(2, 2, L_3) \cap \text{tri}(K_1, K_2)$, up to $4 + L_3$. The quadrupole weight on leg 1 is a rank-2 harmonic of $\hat{r}_1$ in the unprimed kernel and a rank-2 harmonic of $\hat{p}_1$ in the primed one; the anchor carries the two line-of-sight quadrupoles of the estimators and the vertex anisotropy. For

B_{202} the second class-1 pairing is not the mirror of the first ($\ell_1 \neq \ell_2$), so both must be evaluated; the diagonal of B_{202} at $k_1 = k_2$ receives the swapped term with the roles of the two legs exchanged.

Recipe for the diagonal. (i) Pair counts with the anchored moments $A_{K\kappa}^{(a)}(x; r)$ for $\lambda \leq \lambda_w$, $L_a \in \{0, 2, 4\}$, and the eight (W, S) combinations; (ii) the invariants $\mathcal{Q}$ for the labels above, contracted per anchor; (iii) the kernels: $\bar{j}_\ell^{(i)}$ from the shells, the two-dimensional $\mathcal{P}$ of Equation 22 from the model multipoles and the bipolar coefficients $c_{ab}^{L_3}(p_1, p_2)$ (a one-dimensional integral per (a, b, L_3) and per node of the two shells), and for class 2 the projected kernel by quadrature; (iv) the coefficients $\mathcal{T}$ from 3- j symbols; (v) the double radial integral. The class-1 terms give the diagonal to $O(\Delta k/k)$; adding the four class-2 terms completes the Gaussian part, on and off the diagonal.

8 Checks and caveats

- **Box limit.** Both classes reduce to eq. (49) of Sugiyama et al. (2020) when $W = 1$; in the truncated scheme this is the statement that only the isotropic $\mathcal{Q}$ survives, so the box limit tests the kernels, the counting factors and the $\mathcal{T}$ coefficients independently of the pair counts.
- **Gaussian-field test.** As for the power spectrum, Gaussian realisations on the real footprint (`run_gaussian_footprint.py`) measured with the Sugiyama estimator give the exact Gaussian covariance of the windowed multipoles, including the window three-point effects at all ranks, and test λ_w .
- **Which momentum carries the power spectrum.** In Equation 12 to Equation 14 the spectrum is evaluated at the primed momentum; the symmetric choice (the mean over the two shells, or half the sum of the two assignments) differs at $O(\Delta k/k)$ and is the one to use for the final product, as for the power spectrum.
- **Dropped pairings.** The pairings with a pair inside one estimator are the integral-constraint terms of the bispectrum; they matter only for shells at the window scale.
- **Scope.** Gaussian term only: the BB , PT and P_6 terms of the bispectrum covariance (Sugiyama et al. 2020, eqs. 31–33) are not small at $k \gtrsim 0.1$ and need the non-Gaussian machinery of `thecov.trispectrum`; the super-sample term is separate. The window is the local one; a pair-averaged window enters, as for the power spectrum, through the dilution of the mean and through W in Equation 27.